

SAFETY DATA SHEET (GHS)

Acetophenone | CAS: 98-86-2 | Formula: C₈H₈O

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Acetophenone, CAS 98-86-2 (<https://echa.europa.eu/>)
- PubChem CID 7410 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/7410>
- Fisher Scientific Safety Data Sheet — Acetophenone, <https://www.fishersci.com/>
- Wikipedia — Acetophenone (physical property cross-check), <https://en.wikipedia.org/wiki/Acetophenone>
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Acetophenone

CAS Number: 98-86-2

Chemical formula: C₈H₈O

Molecular weight: 120.15 g/mol

Synonyms: Methyl phenyl ketone; 1-phenylethanone; acetylbenzene

Process role: Minor by-product/impurity in cumene-process streams associated with DMBA/AMS chemistry. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Flam. Liq. 4 (H227); Acute Tox. 4 (H302); Eye Irrit. 2A (H319); Aquatic Acute 3 (H402, non-mandatory)

Signal word: Warning

Hazard statements (H-codes): H227 (Combustible liquid); H302 (Harmful if swallowed); H319 (Causes serious eye irritation); H402 (Harmful to aquatic life — non-mandatory classification)

Precautionary statements (P-codes): P264, P270, P280, P301+P312, P305+P351+P338, P330, P337+P313, P501

GHS pictograms: GHS07 (Exclamation mark)

Other hazards: Acetophenone is a combustible liquid with flash point above ambient (~77°C) — relevant fire hazard is mainly under heated process conditions rather than at ambient storage temperature. Mild CNS depressant properties reported at high vapour concentrations (typical of ketones).

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Substance: Acetophenone, CAS 98-86-2, typically >98% purity as encountered in process streams/impurity profiles

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air. Obtain medical attention if symptoms (headache, dizziness) persist.

Skin contact: Wash with plenty of soap and water. Remove contaminated clothing.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical attention (H319).

Ingestion: Rinse mouth. Do not induce vomiting. Obtain medical attention (H302).

Notes to physician: Treat symptomatically; monitor for CNS depression following significant exposure

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, alcohol-resistant foam, water spray (for cooling)

Unsuitable extinguishing media: High-volume water jet on a flowing liquid fire

Specific hazards: Combustible liquid (H227) — flash point ~77°C (closed cup); will burn if heated above flash point and exposed to an ignition source

Special protective equipment for firefighters: Self-contained breathing apparatus (SCBA) and full protective clothing for fires involving this material

Hazardous combustion products: Carbon monoxide, carbon dioxide, smoke

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Avoid skin/eye contact and inhalation of vapours/mist; ensure adequate ventilation; wear appropriate PPE (Section 8)

Environmental precautions: Avoid release to drains/watercourses (harmful to aquatic life, H402)

Containment and cleanup: Contain spill with inert absorbent material; collect into closed, labelled containers for disposal

Reference to other sections: See Section 7 (Handling/Storage) and Section 8 (PPE)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Use in well-ventilated areas; avoid prolonged skin contact and inhalation of vapours; avoid eye contact (H319)

Storage conditions: Store in tightly closed containers in a cool, dry, well-ventilated area away from heat and open flame; note that acetophenone has a melting point near room temperature (~19-20°C) and may partially solidify in cool storage — design heating/tracing as needed for handling

Temperature limits: Keep below flash point (~77°C) during storage and handling where practical

Pressure: Atmospheric storage

Segregation requirements: Segregate from strong oxidisers and strong reducing agents; segregate from strong acids and bases

Special requirements: None beyond general combustible-liquid handling practice

Specific end-use recommendations: Where acetophenone is present as a process impurity, monitor concentration trends as an indicator of upstream DMBA/AMS chemistry performance

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): No specific ACGIH TLV or OSHA PEL established for acetophenone; confirm Thailand OEL if applicable

TWA: Not specifically established

STEL: Not established

Ceiling: Not established

Skin notation: Not assigned

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen

Engineering controls: General ventilation adequate for typical handling; local exhaust for tasks generating vapour/mist

PPE — Respiratory: Not normally required; organic vapour respirator if vapour concentrations are elevated

PPE — Gloves: Chemical-resistant gloves (e.g., nitrile)

PPE — Eyes: Chemical splash goggles (H319)

PPE — Body: Standard work clothing; chemical-resistant apron for bulk transfer

Biological monitoring: Not applicable

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless to pale yellow liquid (may solidify near/below room temperature)

Odour: Sweet, pungent, almond/floral-like odour

Odour threshold: Not established

pH: Not applicable

Melting/Freezing point: 19-20°C

Boiling point (at 1 atm): 202°C

Flash point: 77°C (closed cup)
Evaporation rate: Slower than water
Flammability (solid/gas): Not applicable (liquid at typical process temperatures)
Flammability limits in air: Lower ~1.1% v/v; Upper not well established (confirm against authoritative source)
Vapour pressure (at 20°C): ~0.04 kPa (~0.3 mmHg)
Vapour density (air = 1): 4.14
Relative density (water = 1): 1.028 at 20°C
Solubility (water): ~6.1 g/L at 25°C (slightly soluble)
Partition coefficient (log Kow): ~1.58
Auto-ignition temperature: Not well established — confirm against authoritative source
Decomposition temperature: Not established under normal conditions
Viscosity: ~1.6 mPa·s at 25°C
Explosive properties: Not classified as explosive
Oxidising properties: Not an oxidiser
Molecular weight: 120.15 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions of use and storage
Chemical stability: Stable under recommended storage conditions
Possibility of hazardous reactions: No particular hazardous reactions known under normal process conditions
Conditions to avoid: Heat, open flame, contact with strong oxidisers
Incompatible materials: Strong oxidisers, strong reducing agents (e.g., hydrides), strong acids/bases
Hazardous decomposition products: Carbon monoxide, carbon dioxide on combustion

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Ingestion, inhalation, skin/eye contact
Acute toxicity — Oral LD50: ~815-980 mg/kg, rat (Acute Tox. 4, H302)
Acute toxicity — Dermal LD50: >5,000 mg/kg, rabbit (low acute dermal toxicity)
Acute toxicity — Inhalation LC50: Not well established
Skin corrosion/irritation: May cause mild skin irritation; not classified as corrosive
Serious eye damage/irritation: Causes serious eye irritation (H319)
Respiratory/skin sensitisation: Not classified as a sensitiser
Germ cell mutagenicity: Not classified
Carcinogenicity: Not classified as a carcinogen
Reproductive toxicity: Not classified
STOT (single exposure): Not classified — however, CNS depressant effects (headache, drowsiness) have been reported at high vapour concentrations, typical of ketones
STOT (repeated exposure): Not classified
Aspiration hazard: Not classified

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Harmful to aquatic life (H402, non-mandatory) — fish LC50 generally in the tens-of-mg/L range for acetophenone
Persistence and degradability: Expected to be readily biodegradable
Bioaccumulative potential: Low bioaccumulation potential (log Kow ~1.58)
Mobility in soil: Moderate to high mobility (moderate water solubility)
Other adverse effects: None specifically identified

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste incineration or treatment facility in accordance with local regulations; do not discharge to drains

Contaminated packaging: Dispose of as hazardous waste or, if uncontaminated/empty, per local regulations after appropriate purging

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 (verify applicability); local industrial waste regulations

SECTION 14 — TRANSPORT INFORMATION

UN Number: Not regulated as a dangerous good for transport (acetophenone is generally not assigned a UN number — combustible liquid with flash point >60°C is typically outside Class 3 scope under most transport regulations)

Proper shipping name: Not applicable / Not regulated

Transport hazard class(es): Not classified as a transport hazard (confirm for bulk marine/road transport given combustible liquid status)

Packing group: Not applicable

Environmental hazards (marine pollutant): Not classified as a marine pollutant

Special precautions for user: None specific beyond standard good practice for combustible liquid transfer

Bulk transport (MARPOL/IBC): Confirm IBC Code pollution category for bulk marine transport if applicable

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 98-86-2).

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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